List of Test cases to be Tested in Indigenous PMC Demonstration

S.NO		Features to be Tested	Details	Remarks
1		**Indigenous FPGA based IP core or ASIC**		
2		**Source Code of indigenous IP Core to be shown**		
3		**BC, RT and MT mode operations**		
	1	**BC Mode with MT**	BC-RT, RT-BC, RT-RT, Mode code, Broadcast, Asynchronous messages	Vendors PMC as BC HAL Analyser as RT and MT
		Message scheduling		
		Data reading		
		Data writing		
		Data logging with time stamps - MT function		
		Mode code scheduling		
	2	**RT Mode with MT**	BC-RT, RT-BC, RT-RT, Mode code, Broadcast, Asynchronous messages	HAL Analyser will be the BC Vendors PMC as RT and MT
		Data reading		
		Data writing		
		Data logging with time stamps - MT function		
		Mode code reading and writing		
	3	**MT Mode**	BC-RT, RT-BC, RT-RT, Mode code, Broadcast, Asynchronous messages	HAL Analyser will be the BC Vendors PMC as MT
		Data logging with time stamps		
	4	**BC or RT Mode**		
		Independent configuration of BC or RT Mode	BC-RT, RT-BC, RT-RT, Mode code, Broadcast, Asynchronous messages - Case 1 Vendor PMC is BC	Vendors PMC will be the BC HAL Analyser as RT
			BC-RT, RT-BC, RT-RT, Mode code, Broadcast, Asynchronous messages - Case 2 Vendor PMC is RT	HAL Analyser will be the BC Vendors PMC as RT
	5	**Option for synchronising application software with MIL 1553B scheduling**	The application software should be able to control the scheduling of MIL-1553B messages through general purpose flag(s) or through some other mechanism.	
4		**DMA Operations in BC, RT and MT Modes**		
	1	BC Mode	BC-RT, RT-BC, RT-RT, Mode code, Broadcast, Asynchronous messages	Vendor to verify and validate the time difference between DMA and Non- DMA mode operations
	2	RT Mode	BC-RT, RT-BC, RT-RT, Mode code, Broadcast, Asynchronous messages	
	3	MT Mode	BC-RT, RT-BC, RT-RT, Mode code, Broadcast, Asynchronous messages	
5		**Mode Code Verification both in BC and RT mode**		Detailed verification of mode codes
	1	BC Mode		
	2	RT Mode		
6		**Broadcast mode both in BC and RT mode**		Detailed verification of broadcast messages
	1	BC Mode		
	2	RT Mode		
7		**Asynchronous messages**		Detailed verification of both high and low priority Asynchronous messages
8		**Built In Test – both PBIT and CBIT**		
9		**Bus Load Analysis – Through analyser**		For both Dummy ICD and 100% load
10		**VxWorks 6 or above Compliant APIs**		
11		**Register level programming**		
12		**Dual redundancy and message Retry**		
13		**Timing Analysis** (including Signal levels with respect to timing, inter message gap, Response Time, Response Time out, data loss if any, Maximum No of retries etc.,)		Time stamping using high resolution timers
14		**Check Full words 32**		Some messages should be of 32 words
15		**MIL 1553B ICD to be used for testing**		Enclosed herewith as Annexure - 1

MIL-1553B ICD for testing of

Indigenous MIL-1553B PMC

MISSION & COMBAT SYSTEM RESEARCH & DESIGN CENTER
DESIGN COMPLEX
HINDUSTAN AERONUATICS LIMITED
BANGALORE 560017

ABSTRACT

MIL-1553B Bus Controller functionality implementation for the application software of any avionics system is a very critical and complicated activity. Since MIL-1553B is the communication bus between the Main Avionics Computer and the other LRUs integrated to it, the successful implementation of the Bus Controller functionality is very critical to the application software. The MIL-1553B bus functionality and the other operational functionalities of the application software should meet the scheduling time constraint specified for the avionics system. This document specifies the MIL-1553B messages and the scheduling method to be implemented for testing the indigenous MIL-1553B PMC.

MIL-1553B Frame for a Single Bus

Sl. No.	FRAME#0	FRAME#1	FRAME#2	FRAME#3	FRAME#4	FRAME#5	FRAME#6	FRAME#7
1.	SYNC A	SYNC A	SYNC A	SYNC A	SYNC A	SYNC A	SYNC A	SYNC A
2.	SYNC B	SYNC B	SYNC B	SYNC B	SYNC B	SYNC B	SYNC B	SYNC B
3.	PMC-4-A	-	-	-	-	-	-	-
4.	PMC-4-B	-	-	-	-	-	-	-
5.	PMCX-301	PMCX-301	PMCX-301	PMCX-301	PMCX-301	PMCX-301	PMCX-301	PMCX-301
6.	PMC-108	PMC-108	PMC-108	PMC-108	PMC-108	PMC-108	PMC-108	PMC-108
7.	PMC-109	PMCX-303	PMC-109	PMCX-303	PMC-109	PMCX-303	PMC-109	PMCX-303
8.	PMCX-303	PMC-109	PMCX-303	PMC-109	PMCX-303	PMC-109	PMCX-303	PMC-109
9.	PMCX-110	PMCX-110	PMCX-110	PMCX-110	PMCX-110	PMC-1	SYS05-1	PMCX-110
10.	SYS21-101	PMC-2	PMCX-116	PMCX-116	SYS21-101	PMCX-110	PMCX-110	SYS06-1
11.	SYS22-101	PMC-117	SYS11-101	SYS11-101	SYS22-101	PMCX-116	SYS07-1	PMCX-116
12.	PMCX-116	PMC-121	PMC-103	SYS16-101A	PMCX-116	SYS11-101	PMCX-116	SYS11-101
13.	SYS11-101	PMCX-116	PMCX-302	BIS22-1	SYS11-101	PMC-156	SYS11-101	SYS05-101
14.	PMC1-112	SYS11-101	SYS16-111	SYS07-103	SYS21-106	SYS11-1	PMCX-302	SYS06-105
15.	PMCX-307A	PMC-156	SYS07-102	PMC-157	SYS22-106	SYS21-1	SYS15-1	SYS07-105
16.	SYS05-2	SYS21-104	PMC-155	PMC-1	PMCX-307A	SYS22-1	SYS13-1	PMC-157
17.	PMC-157	SYS22-104	PMCX-307A	PMC2-5	PMC-6	PMCX-306	SYS06-101	PMC-1
18.	BIS2X-101	PMCX-306	SYS05-2	BIS21-1	SYS05-2	PMC-157	PMCX-307A	SYS21-103
19.	BIS22-201	SYS11-102	PMC-157	SYS15-104	PMC-157	PMC-145	SYS07-104	SYS22-103
20.	PMCX-307B	PMC-157	PMC-114	PMC-102	SYS15-111	PMC-111	PMC-155	PMC-106
21.	SYS07-101	PMC-1	SYS22-105	PMCX-304A	SYS15-111A	PMC-130	SYS05-2	PMCX-304A
22.	PMC2-117	PMC-130	SYS21-105	SYS07-101	PMC-200	SYS15-109	PMC-157	PMC-106A
23.	PMC-107	SYS16-101	SYS15-105	PMC-154	PMCX-307B	SYS15-103	SYS16-1	SYS07-101
24.	PMCX-307C	SYS15-101	PMCX-307B	PMC-134	SYS07-101	SYS07-101	SYS16-111A	PMC-154
25.	PMC-113A	SYS06-103	SYS07-101	PMC-120A	PMC-122	PMCX-304A	SYS15-105	PMC-134
26.	PMC-154	PMCX-304A	PMC-154	PMCX-304B	PMC-113A	PMC-154	PMCX-307B	PMCX-304B
27.	BIS2X-301	SYS15-109	PMC-134	PMC-153	PMCX-307C	PMC-134	PMC-118A	PMC-120A
28.	PMC-101	SYS07-101	PMC-120	PMC-132	PMC-154	PMC-120A	PMC1-117	PMC-153
29.	PMC-134	PMC-154	PMCX-307C	PMC-158	PMC-134	PMCX-304B	SYS07-101	PMC-555
30.	PMC-120	PMCX-304B	PMC-153	BIS2X-103	BIS2X-301	PMC-153	PMC-154	PMC-158
31.	PMCX-307D	PMC-134	BIS2X-301	PMC-160	PMC-120	SYS13-6	PMCX-307C	BIS2X-103

Sl. No.	FRAME#0	FRAME#1	FRAME#2	FRAME#3	FRAME#4	FRAME#5	FRAME#6	FRAME#7
32.	PMC-153	PMC-120A	PMC-131	BIS21-4	BIS2X-303	PMC-121	PMC-134	PMC-165
33.	PMC-128	PMC-153	BIS2X-103	SYS11-2	PMC-153	PMC-124	PMC-120	SYS21-107
34.	BIS2X-103	SYS13-6	PMCX-307D	BIS2X-302B/ BIS2X-306A	PMCX-307D	PMC-135	BIS2X-301	SYS22-107
35.	SYS05-3	BIS2X-103	PMCX-309	SYS21-107	PMC-133	BIS2X-103	PMC-153	BIS2X-402
36.	SYS07-2	SYS15-2	PMCX-310	SYS22-107	BIS2X-103	SYS21-107	PMC2-112	PMC-129
37.	PMCX-308A	BIS21-2	PMCX-308A	PMCX-308A	SYS13-2	SYS22-107	PMCX-307D	PMC-162
38.	PMCX-308B	BIS22-2	PMCX-308B	PMCX-308B	PMCX-308A	SYS06-104	SYS10-1	PMC-159
39.	BIS2X-305A	SYS06-3	BIS2X-305A	BIS2X-305A	PMCX-308B	PMCX-308A	BIS2X-103	SYS21-102
40.	BIS2X-305B	SYS10-2	BIS2X-305B	BIS2X-305B	BIS2X-305A	PMCX-308B	PMCX-309	SYS22-102
41.	**PMC2-5**	BIS2X-302A	**PMC2-5**	**PMC-123**	BIS2X-305B	BIS2X-305A	PMCX-310	PMCX-308A
42.	POLL-SYS15-A	SYS21-107	**BIS22-201**	**BIS22-201**	BIS2X-304A	BIS2X-305B	PMCX-308A	PMCX-308B
43.	POLL-SYS15-B	SYS22-107	POLL-SYS05-A	**SYS16-2**	BIS2X-304B	**PMC2-5**	PMCX-308B	BIS2X-305A
44.		PMCX-308A	POLL-SYS11-A	POLL-SYS21-A	**PMC2-5**	**BIS22-201**	BIS2X-305A	BIS2X-305B
45.		PMCX-308B	POLL-SYS05-B	POLL-SYS22-A	**BIS22-201**	POLL-BIS1-A	BIS2X-305B	**SYS15-102**
46.		BIS2X-305A	POLL-SYS11-B	POLL-SYS21-B	POLL-SYS16-A	POLL-BIS2-A	**PMC2-5**	**SYS16-104**
47.		BIS2X-305B		POLL-SYS22-B	POLL-PMC2-A	POLL-BIS1-B	**BIS22-201**	**PMC-123**
48.		**PMC2-5**			POLL-SYS16-B	POLL-BIS2-B	POLL-SYS13-A	**PMC2-5**
49.		**BIS22-201**			POLL-PMC2-B		POLL-SYS10-A	**BIS22-201**
50.		POLL-SYS06-A					POLL-SYS13-B	POLL-SYS07-A
51.		POLL-SYS06-B					POLL-SYS10-B	POLL-SYS07-B

RT ADDRESSES FOR EACH SYSTEM

```
RT_ADDR_SYS02 = 2;            // RT2
RT_ADDR_SYS05 = 5;            // RT5
RT_ADDR_SYS06 = 6;            // RT6
RT_ADDR_SYS07 = 7;            // RT7
RT_ADDR_SYS10 = 10;           // RT10
RT_ADDR_SYS11 = 11;           // RT11
RT_ADDR_SYS13 = 13;           // RT13
RT_ADDR_SYS14 = 14;           // RT14
RT_ADDR_SYS15 = 15;           // RT15
RT_ADDR_SYS16 = 16;           // RT16
RT_ADDR_SYS21 = 21;           // RT21
RT_ADDR_SYS22 = 22;           // RT22
RT_ADDR_SYS25 = 25;           // BIS1- RT25
RT_ADDR_SYS26 = 26;           // BIS2- RT26
```

Details of MIL-1553B Messages

Sl. No.	Message	Message Type	Source	Destination	Sub-address	No. of Words	Message Frequency (Hz)
1.	SYNC A (Bus A)	Mode Code (TX CMD)	PMC	ALL RTs	-	1	50
2.	SYNC B (Bus B)	Mode Code (TX CMD)	PMC	ALL RTs	-	1	50
3.	PMC-4-A (Bus A)	Broadcast	PMC	ALL RTs	29	2	6.25
4.	PMC-4-B (Bus B)	Broadcast	PMC	ALL RTs	29	2	6.25
5.	PMCX-301	BC-RT	PMC	RT-25/RT-26	16	32	50
6.	PMC-108	BC-RT	PMC	RT-21	1	30	50
7.	PMC-109	BC-RT	PMC	RT-22	1	30	50
8.	PMCX-303	BC-RT	PMC	RT-25/RT-26	4	24	50
9.	PMCX-110	BC-RT	PMC	RT-25/RT-26	1	10	50
10.	SYS21-101	RT-BC	RT-21	PMC	1	26	12.5
11.	SYS22-101	RT-BC	RT-22	PMC	1	26	12.5
12.	PMCX-116	BC-RT	PMC	RT-25/RT-26	3	11	50
13.	SYS11-101	RT-BC	RT-11	PMC	1	14	50
14.	PMC1-112	BC-RT	PMC	RT-25	2	5	6.25
15.	PMCX-307A	BC-RT	PMC	RT-25/RT-26	12	25	25
16.	SYS05-2	RT-BC	RT-5	PMC	2	4	25
17.	PMC-157	BC-RT	PMC	RT-7	5	32	50
18.	BIS2X-101	RT-BC	RT-25/RT-26	PMC	1	24	6.25
19.	BIS22-201	RT-BC	RT-26	PMC	5	1	6.25
20.	PMCX-307B	BC-RT	PMC	RT-25/RT-26	12	14	25
21.	SYS07-101	RT-BC	RT-7	PMC	1	7	50
22.	PMC2-117	BC-RT	PMC	RT-26	9	10	6.25
23.	PMC-107	BC-RT	PMC	RT-15	6	4	6.25
24.	PMCX-307C	BC-RT	PMC	RT-25/RT-26	14	14	25
25.	PMC-113A	BC-RT	PMC	RT-15	2	24	12.5
26.	PMC-154	BC-RT	PMC	RT-7	2	29	50
27.	BIS2X-301	RT-BC	RT-25/RT-26	PMC	10	32	25
28.	PMC-101	BC-RT	PMC	RT-5	1	8	6.25

Sl. No.	Message	Message Type	Source	Destination	Sub-address	No. of Words	Message Frequency (Hz)
29.	PMC-134	BC-RT	PMC	RT-10	7	1	50
30.	PMC-120	BC-RT	PMC	RT-21	2	21	25
31.	PMCX-307D	BC-RT	PMC	RT-25/RT-26	15	14	25
32.	PMC-153	BC-RT	PMC	RT-7	1	17	50
33.	PMC-128	BC-RT	PMC	RT-10	1	16	6.25
34.	BIS2X-103	RT-BC	RT-25/RT-26	PMC	2	11	50
35.	SYS05-3	RT-BC	RT-5	PMC	11	4	6.25
36.	SYS07-2	RT-BC	RT-7	PMC	11	4	6.25
37.	PMCX-308A	BC-RT	PMC	RT-25	7	16	50
38.	PMCX-308B	BC-RT	PMC	RT-25	26	16	50
39.	BIS2X-305A	RT-BC	RT-25/RT-26	PMC	18	10	50
40.	BIS2X-305B	RT-BC	RT-25/RT-26	PMC	19	10	50
41.	PMC2-5	BC-RT	PMC	RT-26	5	1	6.25
42.	POLL-SYS15-A (Bus A)	Mode Code (TX CMD)	PMC	RT-15	-	-	6.25
43.	POLL-SYS15-B (Bus B)	Mode Code (TX CMD)	PMC	RT-15	-	-	6.25
44.	PMC-2	BC-RT	PMC	RT-16	3	5	6.25
45.	PMC-117	BC-RT	PMC	RT-15	1	24	6.25
46.	PMC-121	BC-RT	PMC	RT-6	8	27	12.5
47.	PMC-156	BC-RT	PMC	RT-7	4	32	12.5
48.	SYS21-104	RT-BC	RT-21	PMC	6	1	6.25
49.	SYS22-104	RT-BC	RT-22	PMC	6	1	6.25
50.	PMCX-306	BC-RT	PMC	RT-25/RT-26	11	6	12.5
51.	SYS11-102	RT-BC	RT-11	PMC	3	2	6.25
52.	PMC-123	BC-RT	PMC	RT-6	10	10	12.5
53.	PMC-130	BC-RT	PMC	RT-13	7	12	12.5
54.	SYS16-101	RT-BC	RT-16	PMC	2	32	6.25
55.	SYS15-101	RT-BC	RT-15	PMC	1	19	6.25
56.	SYS06-103	RT-BC	RT-6	PMC	15	26	6.25
57.	PMCX-304A	BC-RT	PMC	RT-25/RT-26	8	27	25
58.	SYS15-109	RT-RT	RT-15	RT-13	TX_SA:5,	4	12.5

Sl. No.	Message	Message Type	Source	Destination	Sub-address	No. of Words	Message Frequency (Hz)
					RX_SA:9		
59.	PMCX-304B	BC-RT	PMC	RT-25/RT26	10	27	25
60.	PMC-120A	BC-RT	PMC	RT-22	2	21	25
61.	SYS13-6	RT-RT	RT-13	RT-15	TX_SA:13, RX_SA:3	24	12.5
62.	SYS15-2	RT-BC	RT-15	PMC	11	9	6.25
63.	BIS21-2	RT-BC	RT-25	PMC	11	27	6.25
64.	BIS22-2	RT-BC	RT-26	PMC	11	27	6.25
65.	SYS06-3	RT-BC	RT-6	PMC	11	4	6.25
66.	SYS10-2	RT-BC	RT-10	PMC	11	4	6.25
67.	BIS2X-302A	RT-BC	RT-25/RT-26	PMC	8	32	6.25
68.	SYS21-107	RT-BC	RT-21	PMC	8	2	25
69.	SYS22-107	RT-BC	RT-22	PMC	0	2	25
70.	POLL-SYS06-A (Bus A)	Mode Code (TX CMD)	PMC	RT-6	-	0	6.25
71.	POLL-SYS06-B (Bus B)	Mode Code (TX CMD)	PMC	RT-6	-	0	6.25
72.	PMC-103	BC-RT	PMC	RT-11	1	5	6.25
73.	PMCX-302	BC-RT	PMC	RT-25/RT-26	6	11	12.5
74.	SYS16-111	RT-BC	RT-16	PMC	25	32	6.25
75.	SYS07-102	RT-BC	RT-7	PMC	2	3	6.25
76.	PMC-155	BC-RT	PMC	RT-7	3	32	12.5
77.	PMC-114	BC-RT	PMC	RT-16	1	9	6.25
78.	SYS22-105	RT-BC	RT-22	PMC	7	21	6.25
79.	SYS21-105	RT-BC	RT-21	PMC	7	21	6.25
80.	SYS15-105	RT-RT	RT-15	RT-2	TX_SA:6, RX_SA:11	19	12.5
81.	PMC-131	BC-RT	PMC	RT-10	4	5	6.25
82.	PMCX-309	BC-RT	PMC	RT-25/RT-26	17	30	12.5
83.	PMCX-310	BC-RT	PMC	RT-25/RT-26	18	25	12.5
84.	POLL-SYS05-A (Bus A)	Mode Code (TX CMD)	PMC	RT-5	-	0	6.25

Sl. No.	Message	Message Type	Source	Destination	Sub-address	No. of Words	Message Frequency (Hz)
85.	POLL-SYS11-A (Bus A)	Mode Code (TX CMD)	PMC	RT-11	-	0	6.25
86.	POLL-SYS05-B (Bus B)	Mode Code (TX CMD)	PMC	RT-5	-	0	6.25
87.	POLL-SYS11-B (Bus B)	Mode Code (TX CMD)	PMC	RT-11	-	0	6.25
88.	SYS16-101A	RT-BC	RT-16	PMC	1	32	6.25
89.	BIS22-1	RT-BC	RT-26	PMC	4	32	6.25
90.	SYS07-103	RT-BC	RT-7	PMC	3	32	6.25
91.	BIS21-1	RT-BC	RT-25	PMC	4	32	6.25
92.	SYS15-104	RT-RT	RT-15	RT-2	TX_SA:10, RX_SA:10	32	6.25
93.	PMC-102	BC-RT	PMC	RT-6	1	17	6.25
94.	PMC-132	BC-RT	PMC	RT-10	5	9	6.25
95.	PMC-158	BC-RT	PMC	RT-7	6	30	12.5
96.	PMC-160	BC-RT	PMC	RT-7	8	30	6.25
97.	BIS21-4	RT-BC	RT-25	PMC	3	19	6.25
98.	SYS11-2	RT-BC	RT-11	PMC	11	4	6.25
99.	BIS2X-302B	RT-BC	RT-25/RT-26	PMC	9	32	6.25
100.	PMC-123	BC-RT	PMC	RT-6	10	10	12.5
101.	SYS16-2	RT-BC	RT-16	PMC	11	17	6.25
102.	POLL-SYS21-A	Mode Code (TX CMD)	PMC	RT-21	-	0	6.25
103.	POLL-SYS22-A	Mode Code (TX CMD)	PMC	RT-22	-	0	6.25
104.	POLL-SYS21-B	Mode Code (TX CMD)	PMC	RT-21	-	0	6.25
105.	POLL-SYS22-B	Mode Code (TX CMD)	PMC	RT-22	-	0	6.25
106.	SYS21-106	RT-BC	RT-21	PMC	5	18	6.25
107.	SYS22-106	RT-BC	RT-22	PMC	5	18	6.25
108.	PMC-6	BC-RT	PMC	RT-6	2	30	6.25
109.	SYS15-111	RT-BC	RT-15	PMC	20	32	6.25

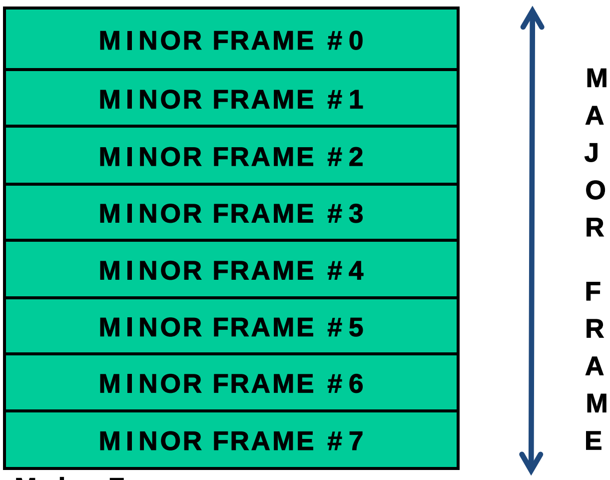
Sl. No.	Message	Message Type	Source	Destination	Sub-address	No. of Words	Message Frequency (Hz)
110.	SYS15-111A	RT-BC	RT-15	PMC	21	32	6.25
111.	PMC-200 (PMC-125)	BC-RT	PMC	RT-5	30	1	6.25
112.	PMC-122	BC-RT	PMC	RT-6	9	17	6.25
113.	BIS2X-303	RT-BC	RT-25/RT-26	PMC	6	32	6.25
114.	PMC-133	BC-RT	PMC	RT-10	6	10	6.25
115.	SYS13-2	RT-BC	RT-13	PMC	11	4	6.25
116.	BIS2X-304A	RT-BC	RT-25/RT-26	PMC	7	16	6.25
117.	BIS2X-304B	RT-BC	RT-25/RT-26	PMC	9	16	6.25
118.	POLL-SYS16-A	Mode Code (TX CMD)	PMC	RT-16	-	0	6.25
119.	POLL-PMC-A	Mode Code (TX CMD)	PMC	RT-14	-	0	6.25
120.	POLL-SYS16-B	Mode Code (TX CMD)	PMC	RT-16	-	0	6.25
121.	POLL-PMC-B	Mode Code (TX CMD)	PMC	RT-14	-	0	6.25
122.	SYS11-1	RT-BC	RT-11	PMC	4	7	6.25
123.	SYS21-1	RT-BC	RT-21	PMC	4	7	6.25
124.	SYS22-1	RT-BC	RT-22	PMC	4	7	6.25
125.	PMC-145	BC-RT	PMC	RT-16	4	2	6.25
126.	PMC-111	BC-RT	PMC	RT-16	2	13	6.25
127.	SYS15-103	RT-RT	RT-15	RT-2	TX_SA:8, RX_SA:8	32	6.25
128.	PMC-124	BC-RT	PMC	RT-6	15	12	6.25
129.	PMC-135	BC-RT	PMC	RT-10	3	14	6.25
130.	SYS06-104	RT-BC	RT-6	PMC	16	22	6.25
131.	POLL-BIS1-A (Bus A)	Mode Code (TX CMD)	PMC	RT-25	-	0	6.25
132.	POLL-BIS2-A (Bus A)	Mode Code (TX CMD)	PMC	RT-26	-	0	6.25
133.	POLL-BIS1-B (Bus B)	Mode Code (TX CMD)	PMC	RT-25	-	0	6.25
134.	POLL-BIS2-B (Bus B)	Mode Code	PMC	RT-26	-	0	6.25

Sl. No.	Message	Message Type	Source	Destination	Sub-address	No. of Words	Message Frequency (Hz)
		(TX CMD)					
135.	SYS05-1	RT-BC	RT-5	PMC	4	6	6.25
136.	SYS07-1	RT-BC	RT-7	PMC	4	7	6.25
137.	SYS15-1	RT-BC	RT-15	PMC	4	20	6.25
138.	SYS13-1	RT-RT	RT-13	RT-15	TX_SA:15, RX_SA:5	25	6.25
139.	SYS06-101	RT-BC	RT-6	PMC	1	16	6.25
140.	SYS07-104	RT-BC	RT-7	PMC	5	32	6.25
141.	SYS16-1	RT-BC	RT-16	PMC	4	20	6.25
142.	SYS16-111A	RT-BC	RT-16	PMC	26	32	6.25
143.	PMC-118A	BC-RT	PMC	RT-15	4	25	6.25
144.	PMC1-117	BC-RT	PMC	RT-25	9	10	6.25
145.	PMC2-112	BC-RT	PMC	RT-26	2	5	6.25
146.	SYS10-1	RT-BC	RT-10	PMC	4	7	6.25
147.	POLL-SYS13-A (Bus A)	Mode Code (TX CMD)	PMC	RT-13	-	0	6.25
148.	POLL-SYS10-A (Bus A)	Mode Code (TX CMD)	PMC	RT-10	-	0	6.25
149.	POLL-SYS13-B (Bus B)	Mode Code (TX CMD)	PMC	RT-13	-	0	6.25
150.	POLL-SYS10-B (Bus B)	Mode Code (TX CMD)	PMC	RT-10	-	0	6.25
151.	SYS06-1	RT-BC	RT-6	PMC	4	9	6.25
152.	SYS05-101	RT-BC	RT-5	PMC	1	7	6.25
153.	SYS06-105	RT-BC	RT-6	PMC	17	6	6.25
154.	SYS07-105	RT-BC	RT-7	PMC	6	28	6.25
155.	SYS21-103	RT-BC	RT-21	PMC	3	12	6.25
156.	SYS22-103	RT-BC	RT-22	PMC	4	12	6.25
157.	PMC-106	BC-RT	PMC	RT-21	6	2	6.25
158.	PMC-106A	BC-RT	PMC	RT-22	6	2	6.25
159.	PMC-555	BC-RT	PMC	RT-10	10	10	6.25
160.	PMC-165	BC-RT	PMC	RT-5	23	4	6.25

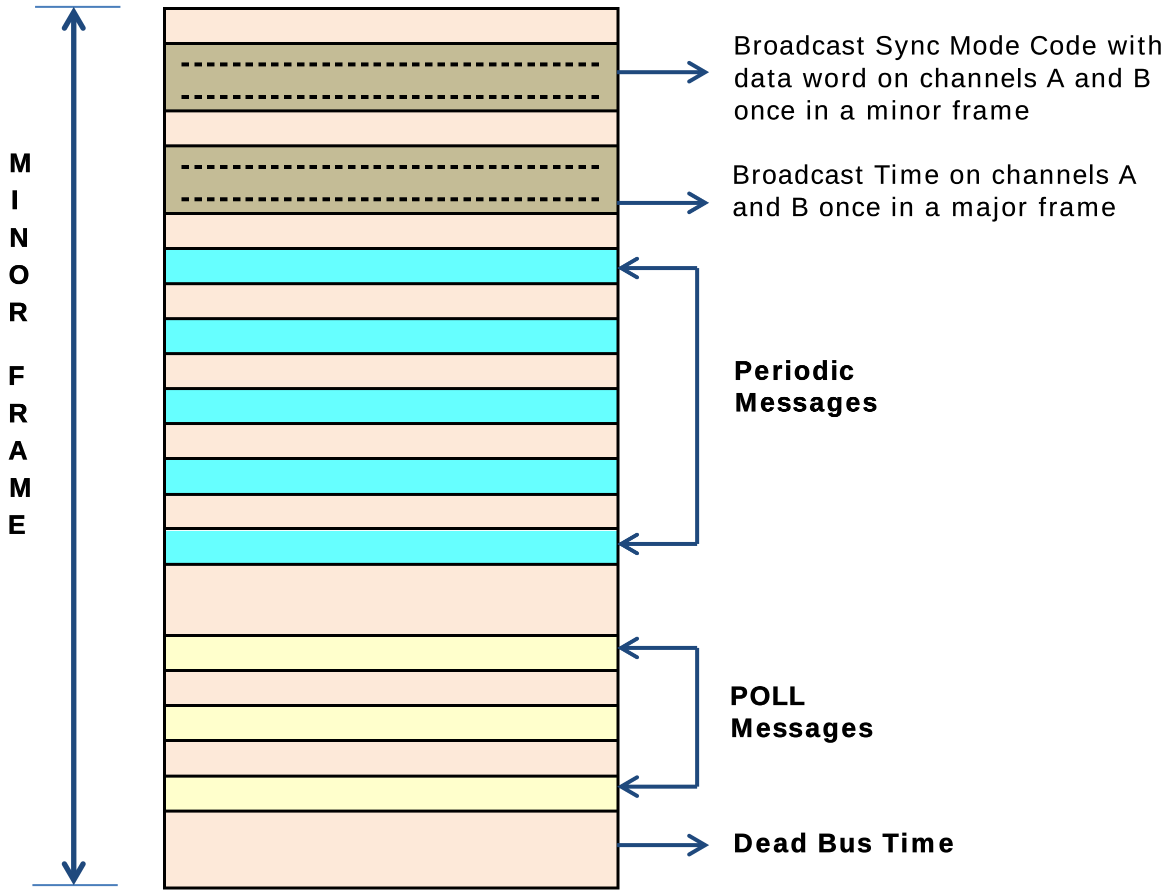
Sl. No.	Message	Message Type	Source	Destination	Sub-address	No. of Words	Message Frequency (Hz)
161.	BIS2X-402	RT-BC	RT-25/RT-26	PMC	14	1	6.25
162.	PMC-129	BC-RT	PMC	RT-10	2	27	6.25
163.	PMC-162	BC-RT	PMC	RT-7	10	14	6.25
164.	PMC-159	BC-RT	PMC	RT-7	7	30	6.25
165.	SYS21-102	RT-RT	RT-21	RT-2	TX_SA:2, RX_SA:15	13	6.25
166.	SYS22-102	RT-RT	RT-22	RT-2	TX_SA:2, RX_SA:16	13	6.25
167.	SYS15-102	RT-BC	RT-15	PMC	3	20	6.25
168.	SYS16-104	RT-BC	RT-16	PMC	3	21	6.25
169.	POLL-SYS07-A (Bus A)	Mode Code (TX CMD)	PMC	RT-7	-	0	6.25
170.	POLL-SYS07-B (Bus B)	Mode Code (TX CMD)	PMC	RT-7	-	0	6.25
171.	BIS2X-306A	RT-BC	RT-25/RT-26	PMC	20	16	6.25
172.	PMC-1	BC-RT	PMC	RT-5	2	32	25

Scheduling Requirements

The PMC should be running the scheduler at 20-milli seconds interval. This duration shall be called the 'minor cycle' and accordingly the bus message frames are formed at this interval as shown in the schedule tables. It is the duration corresponding to the fastest frequency (50Hz) at which the messages are transmitted over the bus (20 m seconds). A 'major cycle' shall be the duration corresponding to the slowest frequency at which the messages are transmitted over the bus (160 m seconds). Each major frame shall contain the 8 minor frames of duration 20 milliseconds. A frame shall comprise all the messages that are to be transmitted over the bus by using the same transfer parameter configuration. It shall include automatically transmitted messages and those transmitted on request.



Major Frame



Minor Frame